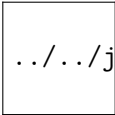


Exam Review: 9.1-9.5

Calculus III | MTH 331 | Section A | Fall 2012

Dr. James Ashe

Johnson C. Smith University



../../jcsu_bull.jpg

Homework

9.1: 21, 55, 60, 66, 68

9.2: 4, 9, 18, 20, 29, 46, 48, 51-56, 58

9.3: 12, 19, 23, 26, 47, 49

9.4: 12, 14, 20, 24, 26, 29

9.5: 9-14, 19-25, 27-28

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$$\lim_{n \rightarrow \infty} \left(\frac{n+x}{n} \right)^n = e^x$$

$$\text{In particular, } \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = e^{-1}.$$

These facts come in useful on problems **9.4:** 29, **9.5:** 12 and 14 (among others)

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Determine the limit of a given sequence.

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- ex.** By using the Squeeze Thm: **ex.** $? \leq \left\{ \frac{n \sin^3 n}{n+1} \right\} \leq ?$.

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- Recognize and evaluate a telescoping series: ex. $\sum_{k=1}^{\infty} \left(\frac{1}{k+2} - \frac{1}{k+3}\right)$ (no method of partial fractions).

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- Use the Divergence Test: ex. $\sum_{k=2}^{\infty} \frac{1}{k \ln k}$.
- Use the Integral Test: ex. $\sum_{k=1}^{\infty} \frac{1}{\sqrt[3]{k+10}}$. These problems may give you the $\frac{d}{dx}$ and $\int dx$.

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You will be provided with all major results